NAND2Tetris Project 3

Sequential Logic and Memory Chips

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Course	NAND2Tetris

1. Bit

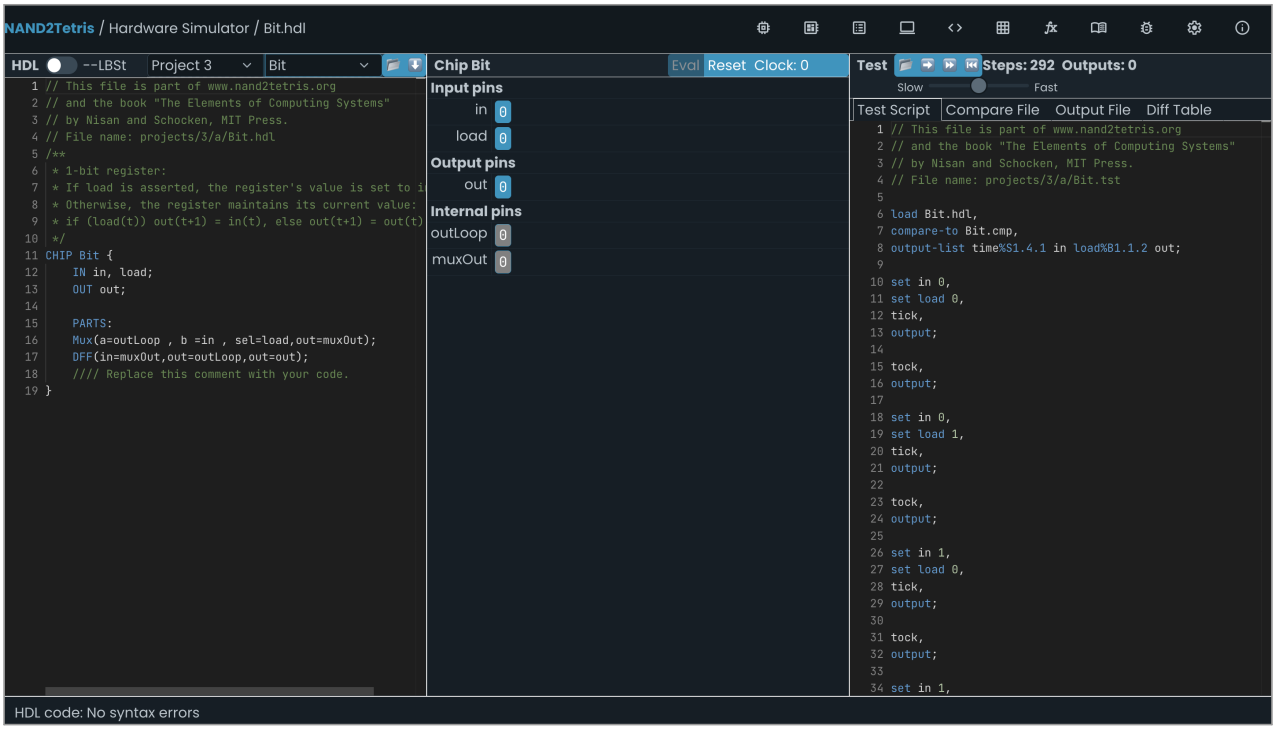


Figure 1: Bit Simulation Successful

```
CHIP Bit {
    IN in, load;
    OUT out;

    PARTS:
    Mux(a=outLoop , b =in , sel=load,out=muxOut);
    DFF(in=muxOut,out=outLoop,out=out);
    //// Replace this comment with your code.
}
```

2. Register

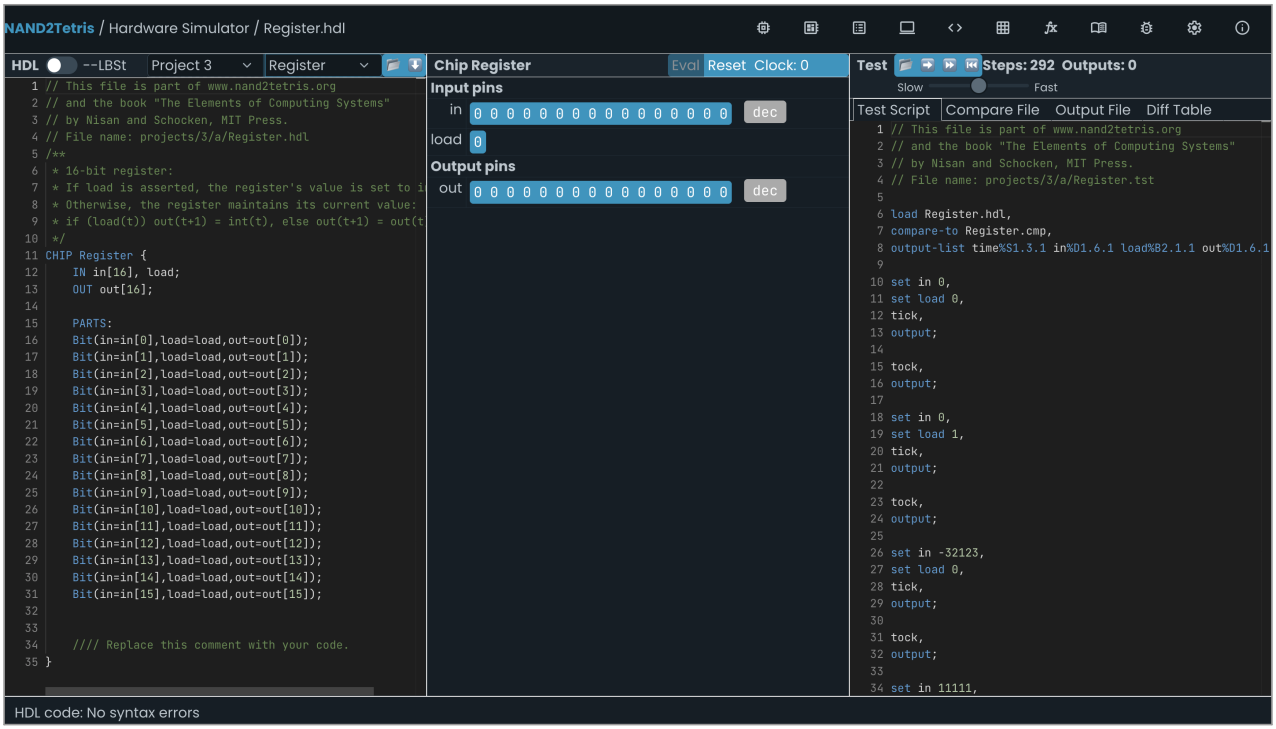


Figure 2: Register Simulation Successful

```
CHIP Register {
    IN in[16], load;
    OUT out[16];

    PARTS:
    Bit(in=in[0],load=load,out=out[0]);
    Bit(in=in[1],load=load,out=out[1]);
    Bit(in=in[2],load=load,out=out[2]);
    Bit(in=in[3],load=load,out=out[3]);
    Bit(in=in[4],load=load,out=out[4]);
    Bit(in=in[5],load=load,out=out[5]);
    Bit(in=in[6],load=load,out=out[6]);
    Bit(in=in[7],load=load,out=out[7]);
    Bit(in=in[8],load=load,out=out[8]);
    Bit(in=in[9],load=load,out=out[9]);
    Bit(in=in[10],load=load,out=out[10]);
    Bit(in=in[11],load=load,out=out[11]);
    Bit(in=in[12],load=load,out=out[12]);
    Bit(in=in[13],load=load,out=out[13]);
    Bit(in=in[14],load=load,out=out[14]);
    Bit(in=in[15],load=load,out=out[15]);


    //// Replace this comment with your code.
}
```

3. RAM8

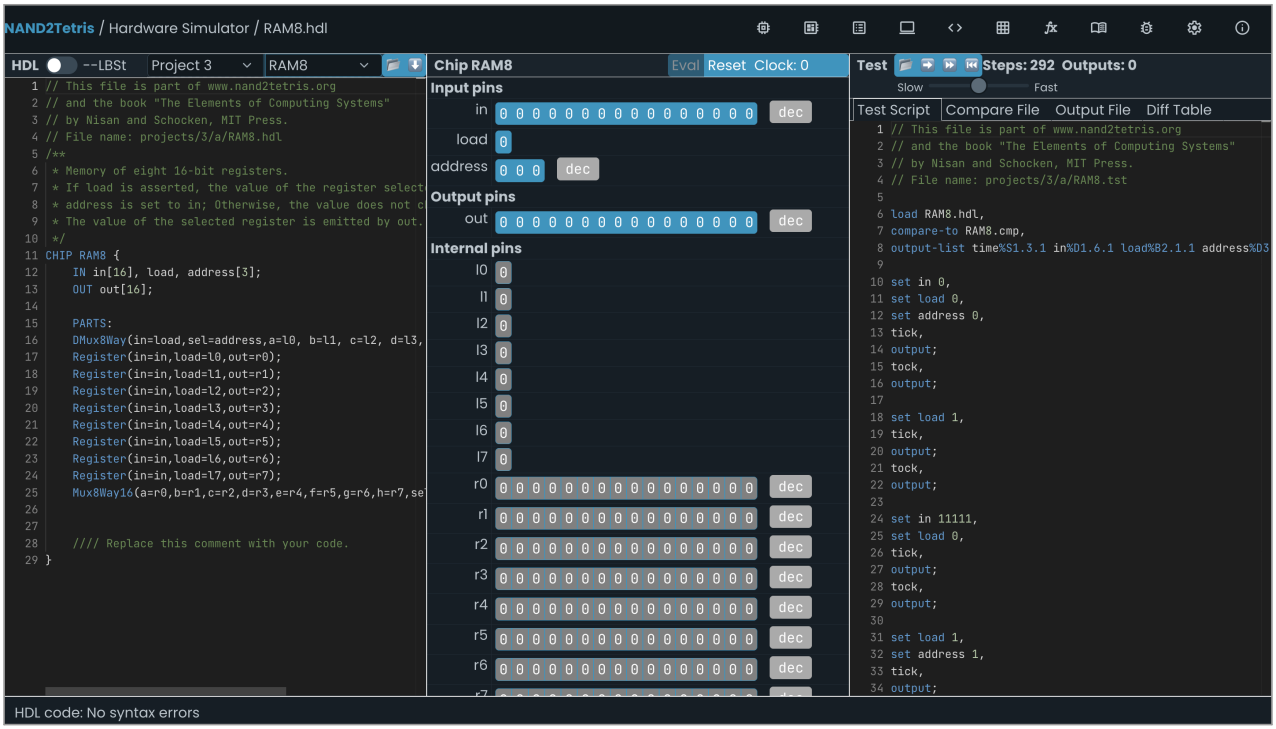


Figure 3: RAM8 Simulation Successful

```
CHIP RAM8 {
    IN in[16], load, address[3];
    OUT out[16];

    PARTS:
    DMux8Way(in=load,sel=address,a=l0, b=l1, c=l2, d=l3, e=l4, f=l5, g=l6, h=l7);
    Register(in=in,load=l0,out=r0);
    Register(in=in,load=l1,out=r1);
    Register(in=in,load=l2,out=r2);
    Register(in=in,load=l3,out=r3);
    Register(in=in,load=l4,out=r4);
    Register(in=in,load=l5,out=r5);
    Register(in=in,load=l6,out=r6);
    Register(in=in,load=l7,out=r7);
    Mux8Way16(a=r0,b=r1,c=r2,d=r3,e=r4,f=r5,g=r6,h=r7,sel=address,out=out);


    //// Replace this comment with your code.
}
```

4. RAM64

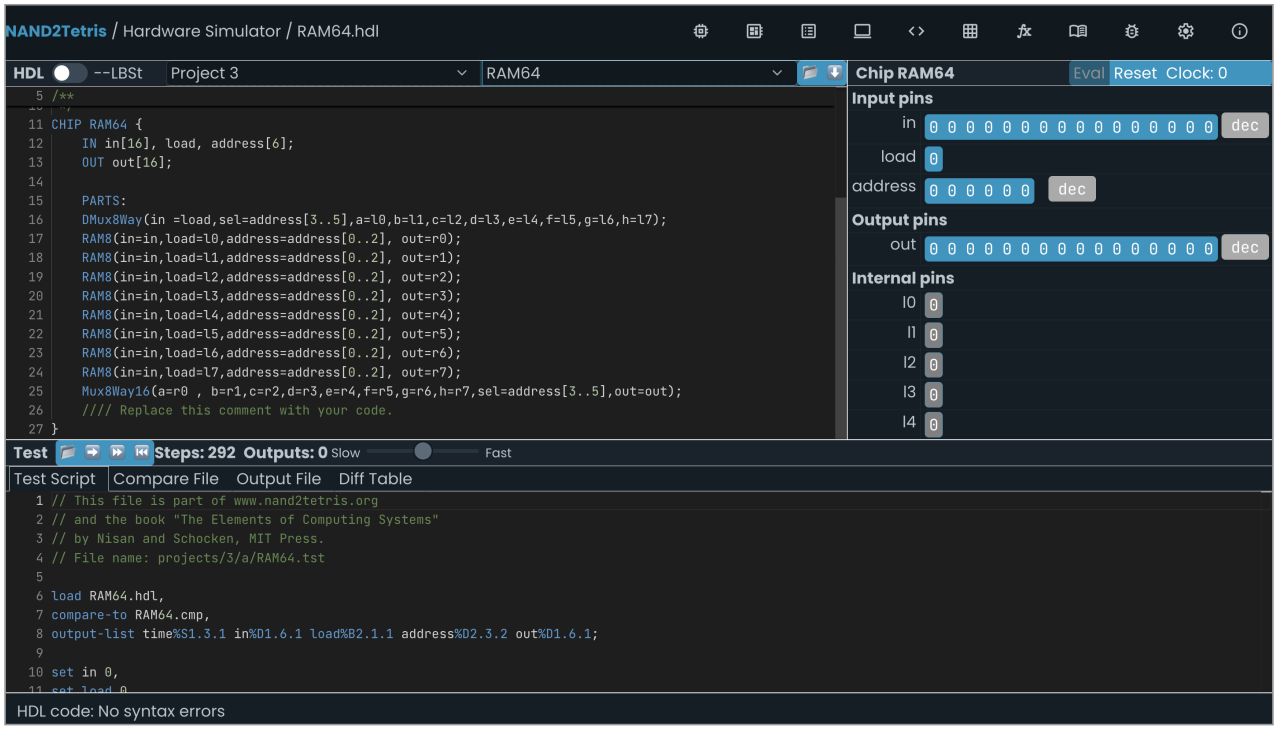


Figure 4: RAM64 Simulation Successful

```
CHIP RAM64 {
    IN in[16], load, address[6];
    OUT out[16];

    PARTS:
    DMux8Way(in =load,sel=address[3..5],a=l0,b=l1,c=l2,d=l3,e=l4,f=l5,g=l6,h=l7);
    RAM8(in=in,load=l0,address=address[0..2], out=r0);
    RAM8(in=in,load=l1,address=address[0..2], out=r1);
    RAM8(in=in,load=l2,address=address[0..2], out=r2);
    RAM8(in=in,load=l3,address=address[0..2], out=r3);
    RAM8(in=in,load=l4,address=address[0..2], out=r4);
    RAM8(in=in,load=l5,address=address[0..2], out=r5);
    RAM8(in=in,load=l6,address=address[0..2], out=r6);
    RAM8(in=in,load=l7,address=address[0..2], out=r7);
    Mux8Way16(a=r0 , b=r1,c=r2,d=r3,e=r4,f=r5,g=r6,h=r7,sel=address[3..5],out=out);
    //// Replace this comment with your code.
}
```

5. RAM512

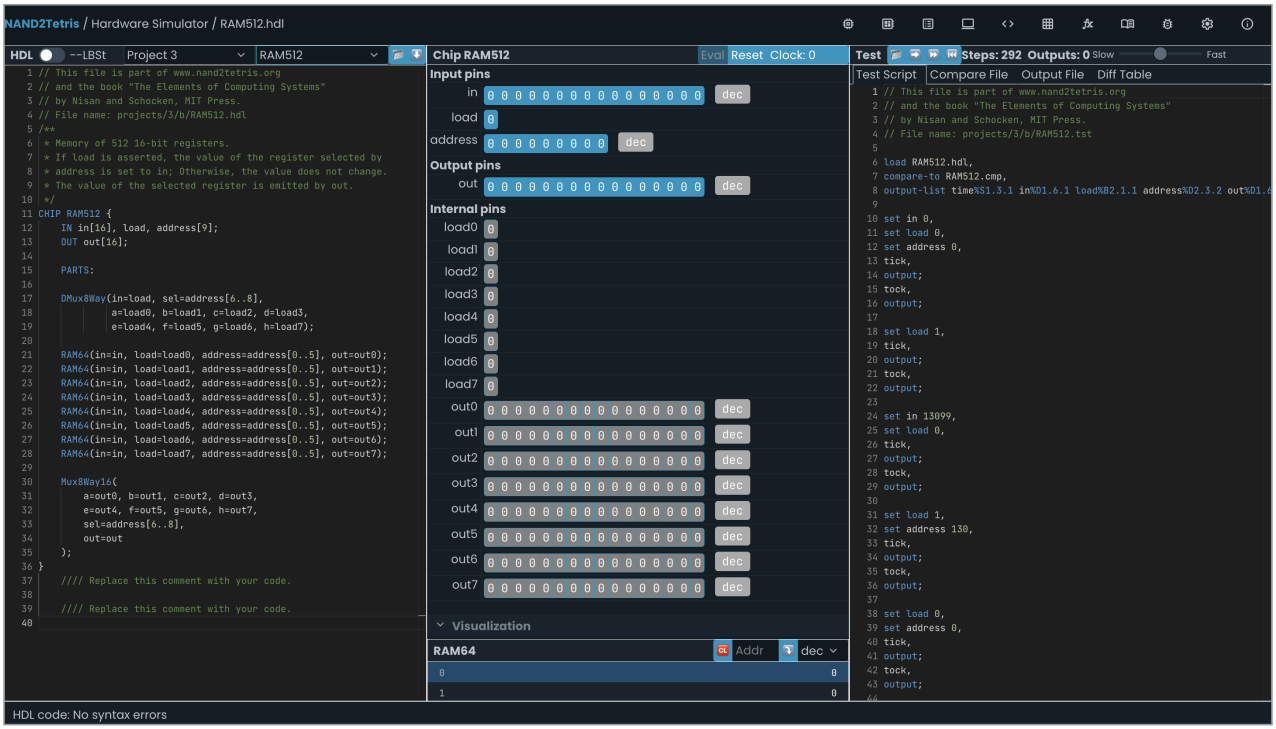


Figure 5: RAM512 Simulation Successful

```
CHIP RAM512 {
    IN in[16], load, address[9];
    OUT out[16];

    PARTS:

    DMux8Way(in=load, sel=address[6..8],
            a=load0, b=load1, c=load2, d=load3,
            e=load4, f=load5, g=load6, h=load7);

    RAM64(in=in, load=load0, address=address[0..5], out=out0);
    RAM64(in=in, load=load1, address=address[0..5], out=out1);
    RAM64(in=in, load=load2, address=address[0..5], out=out2);
    RAM64(in=in, load=load3, address=address[0..5], out=out3);
    RAM64(in=in, load=load4, address=address[0..5], out=out4);
    RAM64(in=in, load=load5, address=address[0..5], out=out5);
    RAM64(in=in, load=load6, address=address[0..5], out=out6);
    RAM64(in=in, load=load7, address=address[0..5], out=out7);

    Mux8Way16(
        a=out0, b=out1, c=out2, d=out3,
        e=out4, f=out5, g=out6, h=out7,
        sel=address[6..8],
        out=out
    );
}
```

6. RAM4K

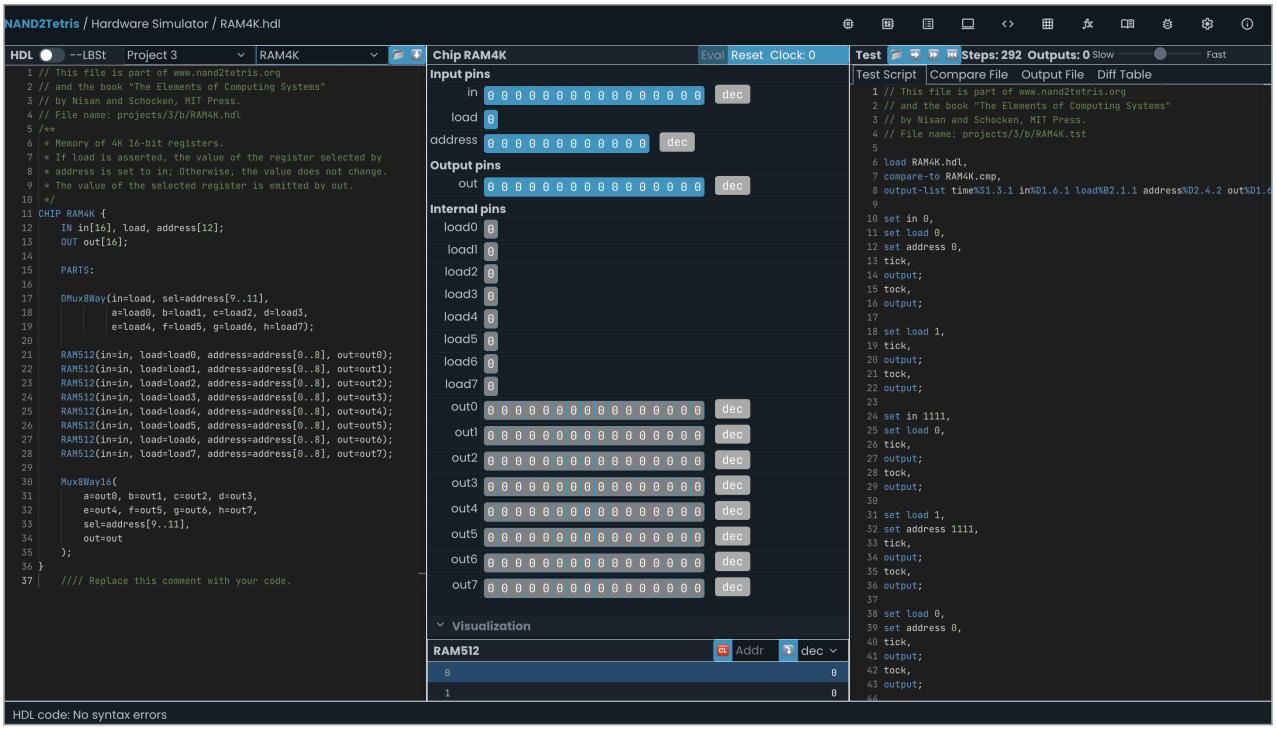


Figure 6: RAM4K Simulation Successful

```
CHIP RAM4K {
    IN in[16], load, address[12];
    OUT out[16];

    PARTS:

    DMux8Way(in=load, sel=address[9..11],
            a=load0, b=load1, c=load2, d=load3,
            e=load4, f=load5, g=load6, h=load7);

    RAM512(in=in, load=load0, address=address[0..8], out=out0);
    RAM512(in=in, load=load1, address=address[0..8], out=out1);
    RAM512(in=in, load=load2, address=address[0..8], out=out2);
    RAM512(in=in, load=load3, address=address[0..8], out=out3);
    RAM512(in=in, load=load4, address=address[0..8], out=out4);
    RAM512(in=in, load=load5, address=address[0..8], out=out5);
    RAM512(in=in, load=load6, address=address[0..8], out=out6);
    RAM512(in=in, load=load7, address=address[0..8], out=out7);

    Mux8Way16(
        a=out0, b=out1, c=out2, d=out3,
        e=out4, f=out5, g=out6, h=out7,
        sel=address[9..11],
        out=out
    );
}
```

7. RAM16K

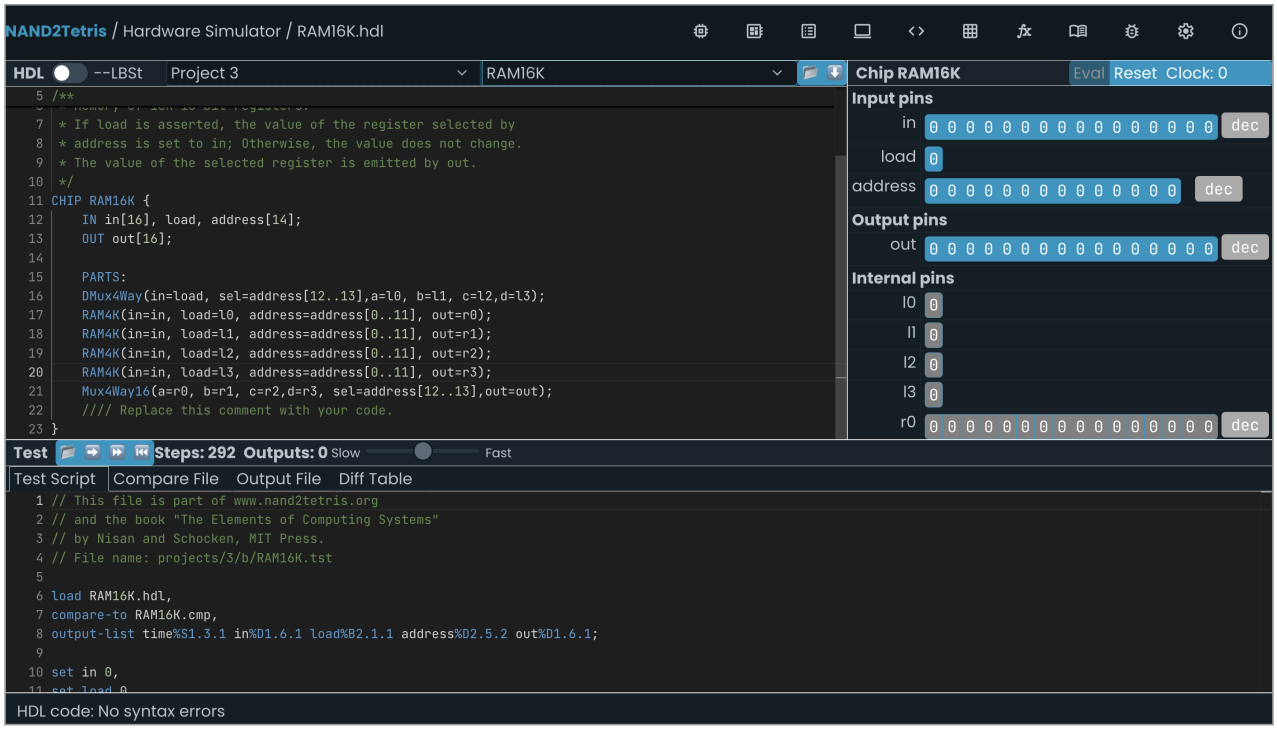


Figure 7: RAM16K Simulation Successful

```
CHIP RAM16K {
    IN in[16], load, address[14];
    OUT out[16];

    PARTS:
    DMux4Way(in=load, sel=address[12..13],a=l0, b=l1, c=l2,d=l3);
    RAM4K(in=in, load=l0, address=address[0..11], out=r0);
    RAM4K(in=in, load=l1, address=address[0..11], out=r1);
    RAM4K(in=in, load=l2, address=address[0..11], out=r2);
    RAM4K(in=in, load=l3, address=address[0..11], out=r3);
    Mux4Way16(a=r0, b=r1, c=r2,d=r3, sel=address[12..13],out=out);
    //// Replace this comment with your code.
}
```

8. PC

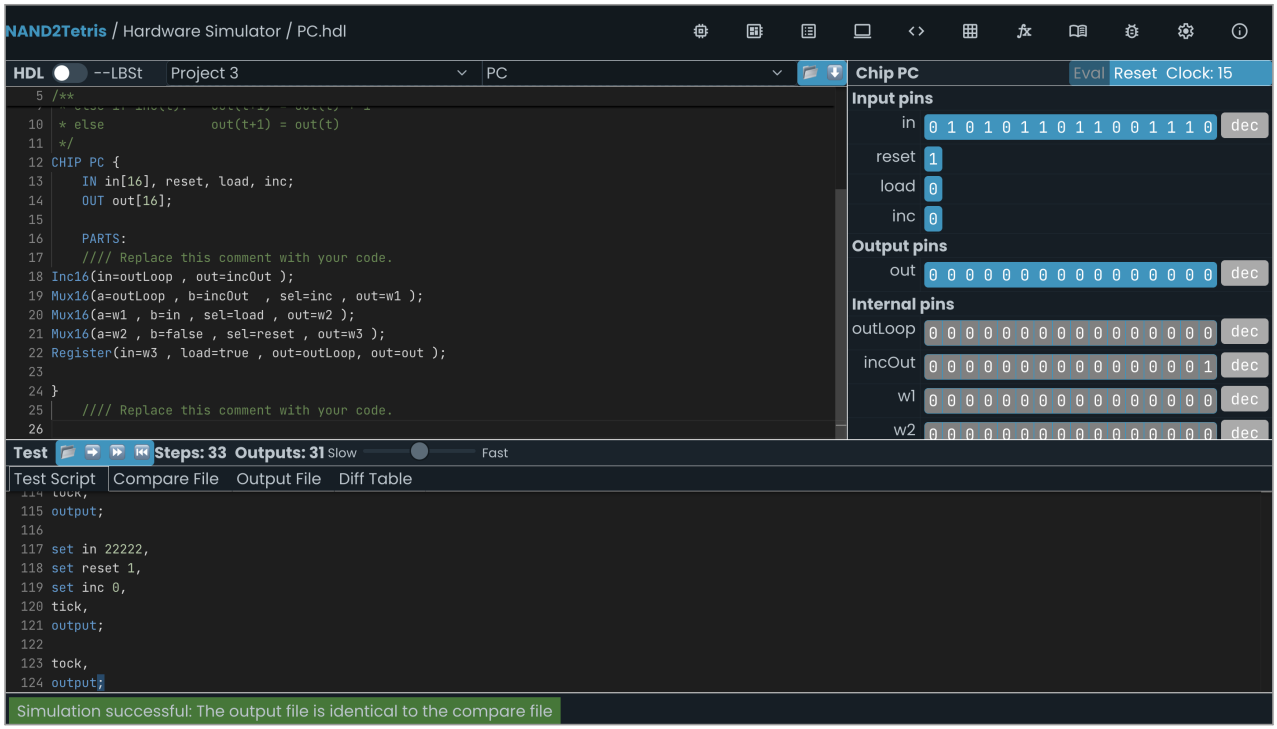


Figure 8: PC Simulation Successful

```
CHIP PC {
    IN in[16], reset, load, inc;
    OUT out[16];

    PARTS:
    //// Replace this comment with your code.
Inc16(in=outLoop , out=incOut );
Mux16(a=outLoop , b=incOut  , sel=inc , out=w1 );
Mux16(a=w1 , b=in , sel=load , out=w2 );
Mux16(a=w2 , b=false , sel=reset , out=w3 );
Register(in=w3 , load=true , out=outLoop, out=out );

}
```